- **LEDs** produce light when current flows from the anode to the cathode. They can be damaged by too much electrical current, so they usually require a protective resistor connecting in series with them.
- LEDs use far less current and thus save energy. They also last much longer as they do not 'burn out'.

Typical mistake

Misunderstanding the difference between input, control and output components and their main functions in a system.

Now test yourself

TESTED

1 A designer is developing an idea for an outdoor security light. The light must come on automatically when it is dark. Name a suitable input sensor and output device for this system. (2)
2 Describe how a transistor operates when being used as an electronic switch. (3)
3 State the main function of a resistor in a circuit. (1)
4 Give **two** applications of buzzers in electronic systems. (2)

Exam tip

Make sure you understand the working characteristics, applications, advantages and disadvantages of the main input, control and output components.

1.7 The use of programmable components

1.7.1 How to make use of flowcharts

REVISED

- Programmable components are used to replace analogue integrated circuits. They can be programmed to perform tasks such as responding to sensors, producing time delays and counting.
- An example is a microcontroller, which has several pins for the connection of different input and output devices.
- Flowchart software is commonly used to program microcontrollers. Standard symbols are joined together using arrows to create a set of instructions for how the system will operate. This is then downloaded on to the chip. Each symbol represents a type of command that can be edited for a specific purpose within the program.
- This type of programming is highly visual, so it is easier to make changes or spot errors. Complex programming knowledge is not necessary. Programs can be simulated on-screen to check that they will work.

Typical mistake

Not using the correct flowchart symbols for each programming command – make sure you know when each of these should be used in a program.

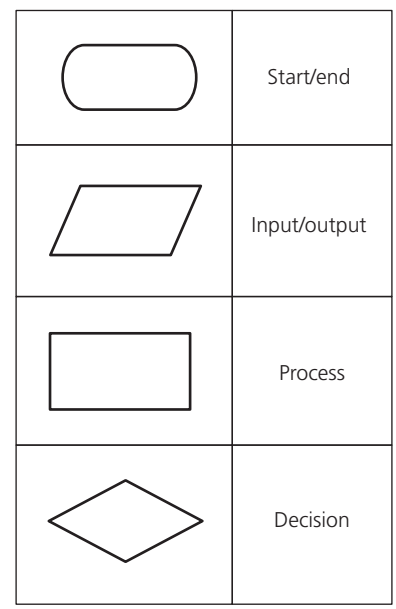


Figure 1.19 Common flowchart symbols

1.7.2 How to switch outputs on/off

REVISED

- In most flowchart software, an output is turned on by inserting the word 'high', followed by the pin number that the output is connected to, into the **input/output** command symbol.
- For example, 'High 2' would turn on the output connected to pin 2. To turn the output off, the word 'low' is used instead.
- **Decision** commands are used to check the state of a digital input device, such as a switch. This is achieved by inserting a question into

the decision command symbol, for example 'is pin 3 on?'. If the answer is yes, the program follows the 'yes' (Y) arrow; if it is no, it follows the 'no' (N) arrow.



Figure 1.20 A command that would turn an output device connected to pin 2 'on'

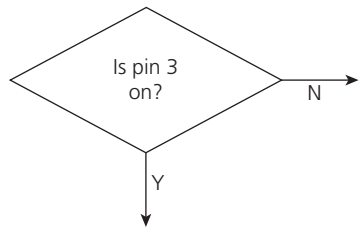


Figure 1.21 A decision command checking whether a digital input device connected to pin 3 is 'on'

Start/End: a command that marks the start or end of a program.

Input/output: a command that is used to change the state of an input or output pin.

Decision: a command that checks the state of a digital input device.

1.7.3 How to process and respond to analogue inputs

REVISED

Many sensors, such as LDRs and thermistors, produce analogue signals. A special type of decision command, called **compare**, can be used to check whether an analogue signal is greater or less than a particular value, known as the threshold.

Compare: a command that checks whether an analogue value is greater or less than the threshold.

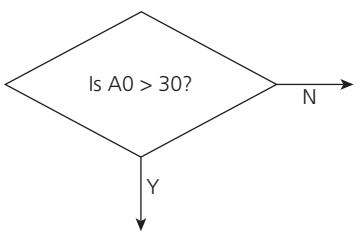


Figure 1.22 A compare command checking whether an analogue input signal is greater than 30

1.7.4 How to use simple routines to control outputs with delays, loops and counts

REVISED

- One of the main reasons for using programmable components is that complex **process** commands can be created with less circuitry, as programming replaces the hardware.
- To create time delays the **wait** command can be used. For example, the program in Figure 1.23 will turn on an output connected to pin 2 when pin 3 detects that a switch has been pressed. The output will remain on for one second and then turn off. The program will then wait another second before looping back to repeat, creating a continuous pulsing sequence.
- Most flowchart programming software has a type of process command that can **count** the number of times an input pin goes from low to high within a certain time period, for example how many times a switch is pressed.

Process: a command that can be used to perform processing functions such as counting or creating time delays.

Wait: a command that is used to create time delays.

Count: a command that can count the number of times an input pin goes from low to high.

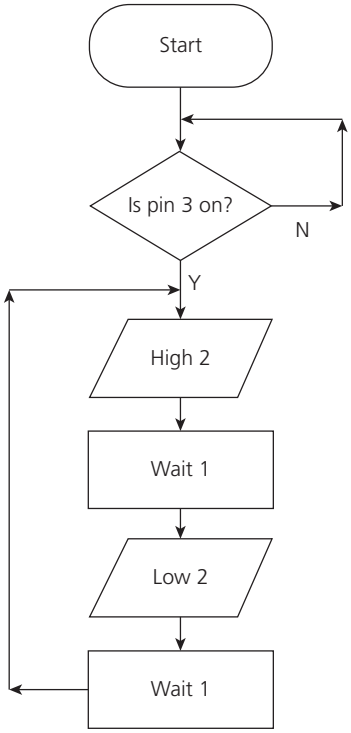


Figure 1.23 A program that uses time delays and loops to create a continuous pulsing sequence

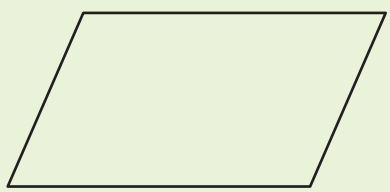
Exam tip

Make sure you can create flowchart programs that can read inputs, switch outputs on/off and produce more complex process functions, such as time delays and counting.

Now test yourself

TESTED

1 What does the following flowchart symbol represent? (1)



2 Draw the flowchart symbol for a process command. (1)

3 Explain how a microcontroller could be programmed to read an analogue value from a sensor. (3)

4 Write a microcontroller flowchart program that would turn an output connected to pin 4 'on' for two seconds when a switch connected to pin 3 is pressed, and then 'off' again. (6)

1.8 Ferrous and non-ferrous metals

1.8.1 Ferrous metals

REVISED

- **Ferrous metals** contain iron so they corrode quickly.
- They are magnetic and corrode quickly if not treated with a suitable surface finish.

Ferrous metals: metals that contain iron.

Mild steel

- Mild steel is also known as low-carbon steel because it has the lowest carbon content of the different steel types.
- Mild steel contains around 99 per cent iron and a maximum of 0.25 per cent carbon.

Assembly drawings

Assembly drawings show how all the different parts of a product fit together. Each part is identified by a number. A bill of materials is included that lists each part needed and the quantities required.

Assembly drawing: drawing that shows how all the different parts of a product fit together, with each part identified by a number.

Schematic diagram: uses symbols to show a circuit in terms of its individual components and how they are connected.

System and schematic diagrams

- System, or block, diagrams show a system in terms of its input, processes and output subsystems. This presents a top-down overview of the system.
- **Schematic diagrams** show a circuit in terms of the individual components used. These are represented using symbols; the connections between them are shown with straight lines.

Computer-aided design

CAD software can be used to produce both drawings and virtual models of designs. Using CAD has many advantages, such as:

- Accuracy of drawing and modelling is very high.
- Changes can be made quickly and easily.
- No materials or physical resources are needed, reducing costs.
- Models can be simulated on screen.
- 2D drawings can be produced directly from the 3D model, saving time.
- CAD files can be outputted directly to CAM equipment for manufacture.

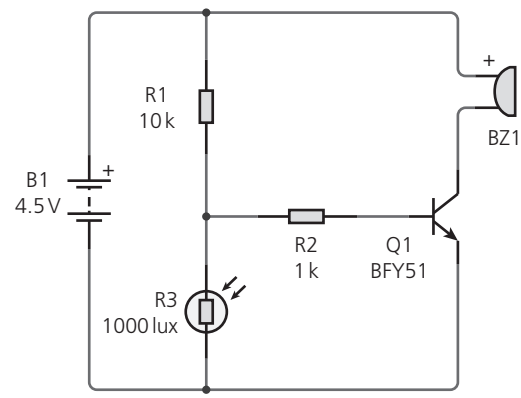


Figure 1.37 An electronic transistor circuit schematic

Exam tip

Make sure you know the purpose and correct use of each of the various ways of developing and communicating design ideas.

1.17.2 Recording and justifying design ideas clearly and effectively

REVISED ☐

- Designers must be able to justify their ideas and the design decisions that they have taken, for example giving reasons why certain materials and components have been chosen. This should always be in terms of the needs and wants of the end user and how well each design would meet them.
- This can be achieved through detailed annotations of design ideas, or listing how each one meets the different requirements of the design specification. Consideration should be given to the wider design context and how each idea fits within this.

Now test yourself

TESTED ☐

1 Describe the differences between freehand sketching and formal drawing. (4)
2 Name **two** types of perspective drawings. (2)
3 Explain the differences between first and third angle orthographic projections. (4)
4 Describe the purpose of a systems diagram. (2)
5 Give **two** reasons why CAD is used to produce drawings of products. (2)